

Probability & Statistics

BCA — Semester III — 2022 — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

11. Differentiate between Convenient sampling and Systematic random sampling with suitable examples.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. From the following data, find Mean and Median. Then use an empirical relation across mean, median and mode to find Mode.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

13. When Mean is 34.69 for 65 total observations, find the missing frequency from the following data.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

14. Use Karl Pearson's correlation method to examine the relationship between marks obtained in two subjects:

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

15. The number of defective articles is found to be 2% in a lot of 300 sample articles per day, then find out following probabilities using suitable probability distribution that(i) There are exactly two defective articles per day(ii) There are at least 2 defective articles per day.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

16. Six coins are tossed 500 times such that probability of getting a head is 21% in each toss. Fit a Binomial distribution for a number of heads with the following frequency distribution:

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

17. In a Hospital, the height of patients follows normal distribution with mean height of 53 inches with standard deviation of 3 inches. Then find out the following probabilities of heights of patients that: i) less than 57 inches, ii) more than 64 inches, iii) between 50 and 56 inches. Use Normal table.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Group C

Attempt any TWO questions[2x10=20]

18. Automobile Company Sells Cars of different model during a year. Data are presented in the following table:

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

19. An analysis of daily wages, paid to workers in two firms A and B belonging to the same industry, gives the following results:

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

20. Perform a one-way ANOVA to differentiate average sales of computers from three electronic shops as given below at 5% level of significance:

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.